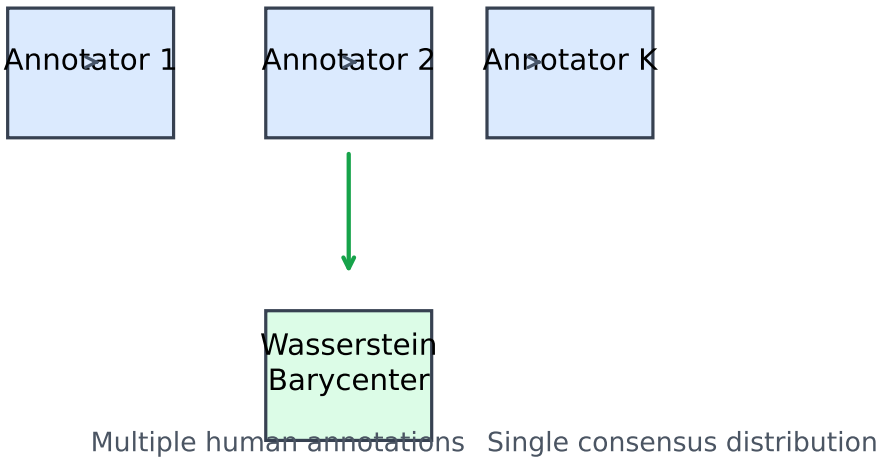
Wasserstein Diffusion on the Annotation Simplex

Distribution-aware consensus modeling for human video summarization



Phase 1 research prototype

IIITK Wasserstein Annotation Project

Problem

Human video summarization naturally gives many annotations per video.

A naive arithmetic mean loses the geometry of disagreement and can blur important saliency peaks.

annotator score curves

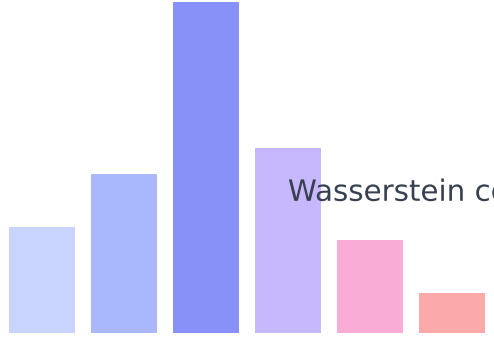
This project models the full annotation distribution on the simplex and aggregates it using optimal transport.

Key idea

Instead of $\bar{x} = \frac{1}{K} \sum_{k=1}^{K} x^{(k)}$, use

$$x^* = \arg \min_{x \in \Delta^{N-1}} \sum_{k=1}^{K} W_2^2(x, x^{(k)})$$

This preserves the transport geometry of the annotation ensemble.



Mathematical formulation

Each annotation is a simplex-valued distribution $x^{(k)} \in \Delta^{N-1}$ with

$$\sum_{i=1}^{N} x_i^{(k)} = 1 \text{ and } x_i^{(k)} \geq 0.$$

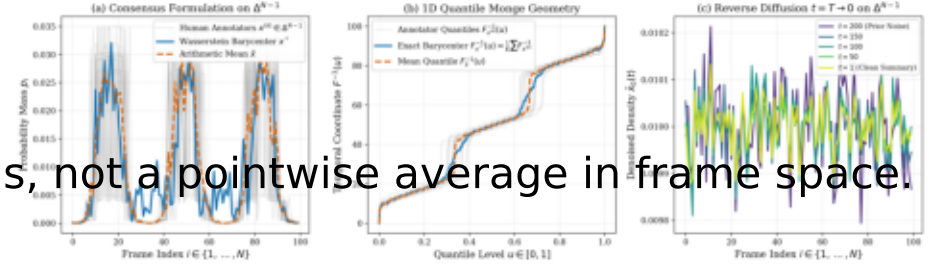
The exact 1D Wasserstein-2 barycenter is defined by

$$x^* = \arg\min_{x \in \Delta^{N-1}} \sum_{k=1}^{K} W_2^2(x, x^{(k)})$$

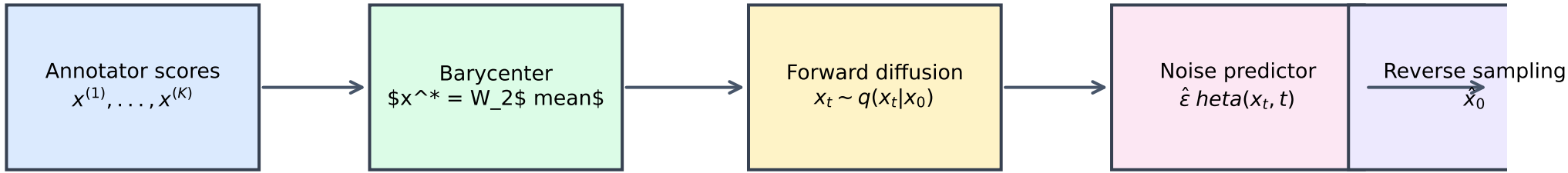
with

$$W_2^2(p, q) = \int_0^1 \left(F_p^{-1}(u) - F_q^{-1}(u)\right)^2 du$$

This is the transport distance between two score distributions, not a pointwise average in frame space.



Model architecture



Distribution-aware consensus target

Training on the simplex

Generative summary output

Training objective

The prototype minimizes a composite objective

$$L = \mathrm{MSE}(\hat{x}_0, x_0) + \lambda \, \mathrm{Cramer}(\hat{x}_0, a)$$

where

$$\mathrm{Cramer}(p, q) = \sum_x (F_p(x) - F_q(x))^2$$

This is practical for training because it is differentiable and matches the CDF structure of the annotation distributions.

The true quantile-based W_2 is used for evaluation, not for direct backpropagation.

This avoids unstable gradients while still measuring the correct downstream consensus quality.

Key results

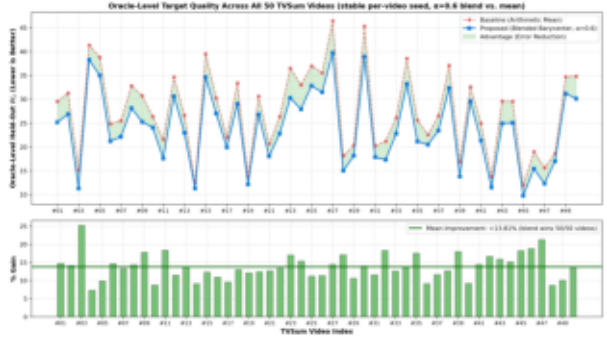
On 50 TVSum videos, the Wasserstein barycenter consistently outperforms the arithmetic mean.

Average held-out W_2 error:

$$26.37 \pm 9.43 \rightarrow 22.01 \pm 8.05$$

Improvement: 16.55% reduction across all videos.

Win rate: 50/50, with a significant paired test result ($p = 1.79 \times 10^{-22}$).



Evaluation protocol

1. TVSum benchmark with 50 videos and 20 annotators per video

2. Leak-free split: 15 annotators used for target construction and training

3. 5 annotators held out and never used for barycenter estimation or backprop

4. After reverse diffusion, generated summaries are scored against the unseen annotators

5. This measures generalization rather than memorization of the training pool

Conclusion

The project treats human annotations as a distribution on the simplex, not as a single average vector.

This enables a more faithful consensus model for video summarization under real disagreement.

The combination of exact 1D Wasserstein barycenters, simplex-aware diffusion, and CDF-based regularization gives more stable and more human-aligned summaries than arithmetic-mean baselines.

Future work: larger transformer denoisers, multimodal conditioning, and general OT formulations beyond the 1D case.

Repository: IIITK Wasserstein Annotation